Ders 9

Z dönüşümünün özellikleri

$x[n] \overset{Z}{\hookrightarrow} X(z)$ ROC: R_x

$x_1[n] \overset{Z}{\hookrightarrow} X_1(z)$ ROC: R_{x_1}

$x_2[n] \overset{Z}{\hookrightarrow} X_2(z)$ ROC: R_{x_2}

$a\,x[n] \overset{Z}{\hookrightarrow} a\,X(z)$ ROC: R_x

$a_1 x_1[n] + a_2 x_2[n] \hookrightarrow a_1 X_1(z) + a_2 X_2(z)$ ROC: $R_{x_1} \cap R_{x_2}$

$x[n-n_0] \overset{Z}{\hookrightarrow} z^{-n_0} X(z)$ ROC: R_x

$z_0^n x[n] \overset{Z}{\hookrightarrow} X(z/z_0)$ ROC: $|z_0| R_x$

Ex: $e^{j\omega_0 n} x[n] \hookrightarrow X(e^{j(\omega-\omega_0)})$

$z_0 = e^{j\omega_0}$

$z_0^n x[n] \hookrightarrow X(z/e^{j\omega_0})$ ROC: R_x

$X(e^{-j\omega_0} z)$

$|e^{j\omega_0}| = 1$

$\angle e^{j\omega_0} = \omega_0$

Ex: $x[n] = r^n \cos(\omega_0 n) u[n]$

$= r^n \left(\frac{e^{j\omega_0 n} + e^{-j\omega_0 n}}{2} \right) u[n]$

$= \frac{1}{2} \underbrace{(r e^{j\omega_0})^n}_{z_1} u[n] + \frac{1}{2} \underbrace{(r e^{-j\omega_0})^n}_{z_2} u[n]$

$\Rightarrow X(z) = \frac{1}{2} \frac{1}{1-z^{-1}}\Big|_{z \leftarrow z/z_1} + \frac{1}{2} \frac{1}{1-z^{-1}}\Big|_{z \leftarrow z/z_2}$

ROC: $|z| > r$

birim adım $u[n] \overset{Z}{\hookrightarrow} \frac{1}{1-z^{-1}}$ ROC: $|z| > 1$

ROC: $|z| > 1 \cdot r$

$n\,x[n] \hookrightarrow -z \frac{dX(z)}{dz}$ ROC: R_x

$u[n]$ $\cos(\omega_0 n)$

$n\,u[n]$ $n \cos(\omega_0 n)$

TABLE 3.1 SOME COMMON z-TRANSFORM PAIRS

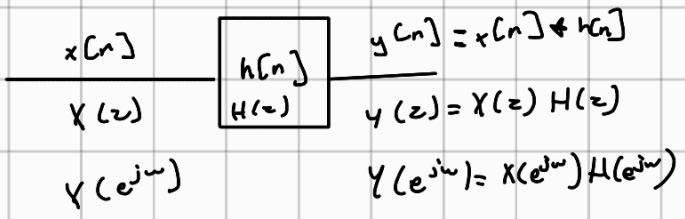
Sequence	Transform	ROC
1. $\delta[n]$	1	All z
2. $u[n]$	$\dfrac{1}{1-z^{-1}}$	$\lvert z\rvert > 1$
3. $-u[-n-1]$	$\dfrac{1}{1-z^{-1}}$	$\lvert z\rvert < 1$
4. $\delta[n-m]$	z^{-m}	All z except 0 (if $m > 0$) or ∞ (if $m < 0$)
5. $a^n u[n]$	$\dfrac{1}{1-az^{-1}}$	$\lvert z\rvert > \lvert a\rvert$
6. $-a^n u[-n-1]$	$\dfrac{1}{1-az^{-1}}$	$\lvert z\rvert < \lvert a\rvert$
7. $na^n u[n]$	$\dfrac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert > \lvert a\rvert$
8. $-na^n u[-n-1]$	$\dfrac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert < \lvert a\rvert$
9. $[\cos\omega_0 n]u[n]$	$\dfrac{1-[\cos\omega_0]z^{-1}}{1-[2\cos\omega_0]z^{-1}+z^{-2}}$	$\lvert z\rvert > 1$
10. $[\sin\omega_0 n]u[n]$	$\dfrac{[\sin\omega_0]z^{-1}}{1-[2\cos\omega_0]z^{-1}+z^{-2}}$	$\lvert z\rvert > 1$
11. $[r^n\cos\omega_0 n]u[n]$	$\dfrac{1-[r\cos\omega_0]z^{-1}}{1-[2r\cos\omega_0]z^{-1}+r^2z^{-2}}$	$\lvert z\rvert > r$
12. $[r^n\sin\omega_0 n]u[n]$	$\dfrac{[r\sin\omega_0]z^{-1}}{1-[2r\cos\omega_0]z^{-1}+r^2z^{-2}}$	$\lvert z\rvert > r$
13. $\begin{cases} a^n, & 0 \le n \le N-1, \\ 0, & \text{otherwise} \end{cases}$	$\dfrac{1-a^Nz^{-N}}{1-az^{-1}}$	$\lvert z\rvert > 0$

TABLE 3.2 SOME z-TRANSFORM PROPERTIES

Section Reference	Sequence	Transform	ROC
	$x[n]$	$X(z)$	R_x
	$x_1[n]$	$X_1(z)$	R_{x_1}
	$x_2[n]$	$X_2(z)$	R_{x_2}
3.4.1	$ax_1[n]+bx_2[n]$	$aX_1(z)+bX_2(z)$	Contains $R_{x_1}\cap R_{x_2}$
3.4.2	$x[n-n_0]$	$z^{-n_0}X(z)$	R_x, except for the possible addition or deletion of the origin or ∞
3.4.3	$z_0^n x[n]$	$X(z/z_0)$	$\lvert z_0\rvert R_x$
3.4.4	$nx[n]$	$-z\dfrac{dX(z)}{dz}$	R_x, except for the possible addition or deletion of the origin or ∞
3.4.5	$x^*[n]$	$X^*(z^*)$	R_x
	$\mathcal{R}e\{x[n]\}$	$\frac{1}{2}[X(z)+X^*(z^*)]$	Contains R_x
	$\mathcal{I}m\{x[n]\}$	$\frac{1}{2j}[X(z)-X^*(z^*)]$	Contains R_x
3.4.6	$x^*[-n]$	$X^*(1/z^*)$	$1/R_x$
3.4.7	$x_1[n]*x_2[n]$	$X_1(z)X_2(z)$	Contains $R_{x_1}\cap R_{x_2}$
3.4.8	Initial-value theorem:		
	$x[n]=0, \quad n<0$	$\lim_{z\to\infty} X(z)=x[0]$	

$x_1[n] * x_2[n] \longleftrightarrow X_1(z)\,X_2(z)$

ROC: $R_{x1} \cap R_{x2}$



† constant coeff. diff equation

$H(z) \rightarrow$ sistem fonksiyonu

$H(s) \rightarrow$ Transfer fonksiyonu

$$x[n] \longleftrightarrow X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$x(z) \longleftrightarrow x[n] = ??$

$$u[n] \longleftrightarrow \frac{1}{1-z^{-1}}$$

1) Tablo yaklaşım yöntemi:

$$x[n] = \left(\frac{1}{2}\right)^n u[n] \xrightarrow{z} X(z) = \frac{1}{1-\frac{1}{2}z^{-1}} \quad |z| > \frac{1}{2}$$

$$x[n] = -\left(\frac{1}{2}\right)^n u[-n-1] \xrightarrow{z} X(z) = \frac{1}{1-\frac{1}{2}z^{-1}} \quad |z| < \frac{1}{2}$$

$$X(z) = \frac{1}{1-\frac{1}{3}z^{-1}} \quad |z| > \frac{1}{3} \longrightarrow x[n] = \left(\frac{1}{3}\right)^n u[n]$$

$$X(z) = \frac{0,5z}{z^2 - z + 0,25} \quad |z| > 0,5$$

$$X(z) = \frac{0,5z}{(z-0,5)^2} = \frac{0,5z^{-1}}{(1-0,5z^{-1})^2}$$

$$x[n] = n\left(\frac{1}{2}\right)^n u[n]$$

z dönüşümü nü bulun

2)

$$X(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}} \qquad \frac{1+2z^{-1}+3z^{-5}}{2+3z^{-4}}$$

$$= \frac{b_0}{a_0} \frac{\prod_{k=0}^{M} (1-c_k z^{-1})}{\prod_{k=1}^{N} (1-d_k z^{-1})} \qquad \frac{(1-3z^{-1})(1-2z^{-1})}{(1-z^{-1})(1-4z^{-1})}$$

$$= \sum_{k=1}^{N} \frac{A_k}{1-d_k z^{-1}} \quad (M<N) \qquad \left(\frac{1}{1-\frac{1}{2}z^{-1}}\right) + \left(\frac{1}{1-\frac{1}{3}z^{-1}}\right) + \cdots$$

$$A_k = (1-d_k z^{-1})\, X(z)\Big|_{z \leftarrow d_k}$$

$$X(z) = \frac{1}{1-\frac{3}{4}z^{-1}+\frac{1}{8}z^{-2}} \qquad |z| > \tfrac{1}{2}$$

$$X(z) = \frac{1}{(1-\frac{1}{4}z^{-1})(1-\frac{1}{2}z^{-1})}$$

$$= \frac{A_1}{1-\frac{1}{4}z^{-1}} + \frac{A_2}{1-\frac{1}{2}z^{-1}} \qquad \subset 2$$

$$A_1 = (1-\tfrac{1}{4}z^{-1})\, X(z)\Big|_{z \leftarrow \frac{1}{4}} = \frac{1}{1-\frac{1}{2}z^{-1}}\Big|_{z \leftarrow \frac{1}{4}} = \frac{1}{1-2} = -1$$

$$A_2 = \frac{1}{1-\frac{1}{4}z^{-1}}\Big|_{z \leftarrow \frac{1}{2}} = 2$$

$$x[n] = -\left(\frac{1}{4}\right)^n u[n] + 2\left(\frac{1}{2}\right)^n u[n]$$

$|z| < \frac{1}{2}$ olsaydı,

$-\left(\frac{1}{4}\right)^n u[-n-1] \Leftrightarrow \frac{1}{1-az^{-1}}$ $|z|<a$

$+\left(\frac{1}{4}\right)^n u[-n-1]$

$-\left(\frac{1}{2}\right)^n u[-n-1]$

$$X(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

$$= \frac{b_0}{a_0} \frac{\prod_{k=0}^{M} (1 - c_k z^{-1})}{\prod_{k=1}^{N} (1 - d_k z^{-1})}$$

$$= \sum_{k=1}^{N} \frac{A_k}{1 - d_k z^{-1}} \quad (M < N)$$

$$= \sum_{r=0}^{M-N} B_r z^{-r} + \sum_{k=1}^{N} \frac{A_k}{1 - d_k z^{-1}} \quad (M \geq N)$$

$B_0 + B_1 z^{-1} + B_2 z^{-2}$

$$\frac{1 + 2z^{-1} + 3z^{-2}}{1 + z^{-1}} = \frac{z^2 + 2z + 3}{z^2 + z} \qquad M = 2,\ N = 1$$

$$X(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 - \frac{3}{2} z^{-1} + \frac{1}{2} z^{-2}} \qquad |z| > 1$$

$M = 2 \quad N = 2 \quad M = N$

$$= 2 + \underbrace{\frac{-1 + 5z^{-1}}{1 - \frac{3}{2} z^{-1} + \frac{1}{2} z^{-2}}}_{M < N}$$

$$\frac{1}{2} z^{-2} - \frac{3}{2} z^{-1} + 1 \;\Big|\; 2$$

$$z^{-2} + 2z^{-1} + 1$$
$$- \; z^{-2} - 3z^{-1} + 2$$
$$\overline{5z^{-1} - 1}$$

$$2 + \frac{-1 + 5z^{-1}}{(1 - z^{-1})\left(1 - \frac{1}{2} z^{-1}\right)} = 2 + \frac{-3}{1 - \frac{1}{2} z^{-1}} + \frac{8}{1 - z^{-1}}$$

$$\Rightarrow x[n] = 2\delta[n] - 3\left(\frac{1}{2}\right)^n u[n] + 8u[n]$$

Çok katlı kök için

$$X(z) = \sum_{r=0}^{M-N} B_r z^{-r} + \sum_{k=1}^{N} \frac{A_k}{1-d_k z^{-1}} + \sum_{m=1}^{s} \frac{C_m}{(1-d_i z^{-1})^m}$$

(s: katlı)

$$C_m = \frac{1}{(s-m)!\,(-d_i)^{s-m}} \left\{ \frac{d^{s-m}}{dw^{s-m}} \left[(1-d_i w)^s X(w^{-1}) \right] \right\}_{w=d_i^{-1}}$$

$z^{-1} = w$

$$Y(z) = \frac{-11+16z^{-1}-7z^{-2}}{(1-\frac{1}{2}z^{-1})^2(1-\frac{1}{3}z^{-1})} \qquad |z| > \frac{1}{2}$$

$$\frac{1}{(2-1)\,(-d_i)^{2-1}} \left[\frac{d}{dw} \left[(-1-d_i w)^2 X(w^{-1}) \right] \right]_{w=d_i^{-1}}$$

$\frac{1}{2}$'de ik: katlı kökümüz

$$= \frac{A_1}{1-\frac{1}{3}z^{-1}} + \frac{C_1}{(1-\frac{1}{2}z^{-1})} + \frac{C_2}{(1-\frac{1}{2}z^{-1})^2}$$

$s=2$, $m=1$

$$C_1 = \frac{1}{1!\,(-\frac{1}{2})^1} \left\{ \frac{d}{dw} \left[(1-\frac{1}{2}w)^2 \cdot \frac{-11+16w-7w^2}{(1-\frac{1}{2}w)^2(1-\frac{1}{3}w)} \right] \right\}_{w=2}$$

$= 114$

$s=2$
$m=2$

$$C_2 = \left. \frac{-11+16w-7w^2}{1-\frac{1}{3}w} \right|_{w=2} = -21$$

$$\rightarrow x[n] = -104\left(\frac{1}{3}\right)^n u[n] + 114\left(\frac{1}{2}\right)^n u[n] - 21(n+1)\left(\frac{1}{2}\right)^{n+1} u[n+1]$$

Tablodaki bu var $\left[\frac{-21z^{-1}}{(1-\frac{1}{2}z^{-1})^2} \right]$ 2. ile çarpılmış $= \frac{-21}{(1-\frac{1}{2}z^{-1})^2}$

$$x[n] \longleftrightarrow X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$X(z) = \sum_{n=-\infty}^{\infty} a_n z^{-n}$$

$$X(z) = z^2 - \frac{1}{2}z - 1 + \frac{1}{2}z^{-1}$$

$d_{-3} = 0$

$x[-2] = a_{-2} = 1$

$x[-1] = a_{-1} = -\frac{1}{2}$

$x[0] = a_0 = -1$

$x[1] = a_1 = \frac{1}{2}$

$x[2] = a_2 = 0$

$$x[n] = \delta[n+2] - \frac{1}{2}\delta[n+1] - \delta[n] + \frac{1}{2}\delta[n-1]$$

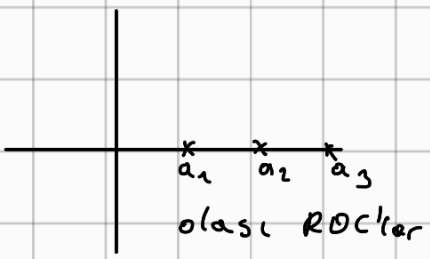
$$x = [1 \; -\frac{1}{2} \; -1 \; \frac{1}{2}]$$

$$X(z) = \frac{1}{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$$

$$(1 - z^{-1})\ (1 - \tfrac{1}{2}z^{-1})$$

$|z| > 1$

$|z| < \frac{1}{2}$



olası ROC'ler

sağ yanlı dizi için $n > 0$

$|z| > 1$ durumu $1 + \frac{3}{2}z^{-1} + \frac{7}{4}z^{-2} + \frac{15}{8}z^{-3} + \cdots$

sol yanlı

$|z| < \frac{1}{2}$

$= 2z^2 + 6z^3 + 14z^4 + 30z^5 + \cdots$

Peki nasıl bulacağız